FINRAG AI

Engineering Main Project Review–1
50 Reviewer / Viva Questions with Answers

PROJECT	FinRAG AI — Integrated Financial Intelligence Platform
CORE	Financial RAG + Transaction Intelligence
PREPARATION	Literature Survey • Research Gap • Novelty • Methodology • Evaluation

How to use this PDF: First memorize the bold technical terms. Then practice answering each question aloud in 20–40 seconds. Do not claim target metrics as achieved results unless you have experimentally measured them.

SECTION 1 — PROJECT BASICS

1. What is the title of your project?

Answer: Our project is **FinRAG AI — an Integrated Financial Intelligence Platform**. It combines Financial RAG for document-based question answering with Transaction Intelligence for transaction-risk analysis.

2. What is FinRAG?

Answer: FinRAG stands for **Financial Retrieval-Augmented Generation**. It is a domain-specific RAG system designed to retrieve relevant financial evidence and use an LLM to generate grounded financial answers.

3. What problem does your project solve?

Answer: Financial documents are lengthy, technical and difficult to search manually. Traditional search may not understand complex financial questions, while standalone LLMs may hallucinate. Our system aims to retrieve relevant evidence and generate traceable, grounded answers.

4. What is the main objective of your project?

Answer: The main objective is to build an AI-powered financial intelligence platform that can retrieve financial information, answer complex questions with evidence and citations, handle financial tables and multi-hop queries, and support transaction-risk analysis.

5. Who are the target users?

Answer: The target users include **financial analysts, compliance officers, investment advisors and retail investors**.

SECTION 2 — LITERATURE SURVEY & BASE PAPER

6. What is the title of your base paper?

Answer: The title is **“Metadata-Driven Retrieval-Augmented Generation for Financial Question Answering.”**

7. Who are the authors of the base paper?

Answer: The authors are **Michail Dadopoulos, Anestis Ladas, Stratos Moschidis and Ioannis Negkakis**.

8. When was the base paper published?

Answer: It is a **2025 arXiv preprint**, with arXiv identifier **2510.24402**.

9. What is the main objective of the base paper?

Answer: The base paper aims to improve financial question answering using a **metadata-driven, multi-stage Retrieval-Augmented Generation architecture**.

10. What is the main contribution of the base paper?

Answer: Its main contribution is using **LLM-generated document and chunk metadata, pre-retrieval filtering, metadata-enriched embeddings and post-retrieval reranking** to improve financial information retrieval.

SECTION 3 — DATASETS

11. Which datasets or sources are used?

Answer: The major sources are **FinanceBench, FinDER and SEC EDGAR financial filings**.

12. What is FinanceBench?

Answer: FinanceBench is a **financial question-answering benchmark** containing more than 1,000 financial QA pairs. It can be used to evaluate the accuracy of financial QA systems.

13. What is FinDER?

Answer: FinDER is a financial dataset containing **5,703 expert-annotated query-evidence-answer triplets**. It is useful for evaluating financial retrieval and evidence-grounded answering.

14. What is SEC EDGAR?

Answer: SEC EDGAR is the electronic filing system of the **U.S. Securities and Exchange Commission**. It provides public corporate filings such as 10-K and 10-Q reports.

15. What are 10-K and 10-Q?

Answer: A **10-K** is an annual corporate filing, while a **10-Q** is a quarterly corporate filing submitted to the SEC.

SECTION 4 — RAG FUNDAMENTALS

16. What is RAG?

Answer: RAG stands for **Retrieval-Augmented Generation**. It retrieves relevant information from an external knowledge source and provides that information as context to an LLM for generating an answer.

17. Why do you need RAG?

Answer: RAG allows the system to use external financial documents as a knowledge source. This helps reduce dependence on information stored inside the LLM and improves evidence grounding.

18. Why use Financial RAG instead of normal RAG?

Answer: Financial documents contain specialized terminology, numerical information, tables and cross-references. Therefore, financial RAG needs specialized retrieval, table handling and financial-document grounding.

19. Why not directly use ChatGPT or another LLM?

Answer: A standalone LLM may generate unsupported information or hallucinate. Our RAG system first retrieves relevant financial evidence and then generates the answer using that evidence.

20. What is hallucination?

Answer: Hallucination occurs when an LLM generates information that is incorrect or unsupported by the available evidence.

SECTION 5 — RAG PIPELINE

21. Explain your RAG pipeline.

Answer: Our pipeline is: **Financial Documents → Parsing → Chunking → Metadata Generation → Embeddings → Indexing → Query Processing → Hybrid Retrieval → Reranking → LLM Generation → Cited Answer.**

22. What is document chunking?

Answer: Document chunking means dividing a large document into smaller meaningful sections so that relevant information can be retrieved efficiently.

23. Why is metadata important?

Answer: Metadata provides additional context about each document or chunk, such as company, reporting period, section or page. It helps the system filter and retrieve more relevant information.

24. What is an embedding?

Answer: An embedding is a numerical vector representation of text that captures its semantic meaning. It allows the system to perform semantic similarity search.

25. What is a vector database?

Answer: A vector database stores embeddings and allows efficient similarity-based retrieval of relevant document chunks.

SECTION 6 — RETRIEVAL & RERANKING

26. What is hybrid retrieval?

Answer: Hybrid retrieval combines **dense semantic retrieval and sparse keyword retrieval such as BM25**. Our project uses this combination to improve retrieval robustness.

27. What is BM25?

Answer: BM25 is a sparse information-retrieval algorithm that ranks documents based on the relevance of query terms.

28. Why combine BM25 with dense retrieval?

Answer: Dense retrieval captures semantic similarity, while BM25 is effective for exact keywords, company names and financial terminology. Combining both provides stronger retrieval.

29. What is reranking?

Answer: Reranking takes the initially retrieved candidate chunks and reorders them according to their relevance to the user's query.

30. Why do you need a cross-encoder reranker?

Answer: A cross-encoder evaluates the query and candidate text together and produces a relevance score. This helps select the most relevant evidence before sending it to the LLM.

SECTION 7 — RESEARCH GAP & NOVELTY

31. What are the limitations of the base paper?

Answer: The identified limitations include **limited evaluation, lack of explicit multi-hop reasoning, text-oriented table handling, static document ingestion, limited citation precision, lack of Indian-market support and latency concerns**.

32. What is the research gap in your project?

Answer: The research gap is the lack of a financial RAG system that comprehensively addresses **table-aware retrieval, multi-hop reasoning, real-time ingestion, precise citations, Indian financial data and efficient response time**.

33. What is the novelty of your proposed system?

Answer: Our proposed extensions include **table-aware chunking, query decomposition and multi-hop reasoning, Indian-market data sources, real-time ingestion, precise page/section/table citations, HyDE and performance optimization**.

34. How is your project different from the base paper?

Answer: The base paper focuses on metadata-driven financial RAG. Our project extends it with **table-aware retrieval, multi-hop reasoning, Indian-market support, real-time ingestion, precise citations and latency optimization**.

35. Why did you select this base paper?

Answer: We selected it because it directly addresses **financial question answering using RAG** and provides a strong technical foundation that we can extend to solve additional limitations.

SECTION 8 — ADVANCED RAG FEATURES

36. What is HyDE?

Answer: HyDE stands for **Hypothetical Document Embeddings**. An LLM generates a hypothetical document-like response to the query, which can then be used to improve semantic retrieval.

37. What is table-aware chunking?

Answer: Table-aware chunking preserves the relationships between **table headers, rows, columns and values** instead of treating a financial table as ordinary text.

38. Why is table-aware retrieval important?

Answer: Financial statements contain important numerical information in tables. If table structure is lost during extraction, the system may retrieve a value without correctly understanding its metric or reporting period.

39. What is multi-hop reasoning?

Answer: Multi-hop reasoning means answering a question that requires information from multiple pieces of evidence or multiple reasoning steps.

40. How will you handle a multi-hop question?

Answer: We propose **query decomposition**. A complex question is divided into smaller sub-questions, evidence is retrieved for each step, and the intermediate results are combined to generate the final grounded answer.

SECTION 9 — INDIAN MARKET & TRANSACTION INTELLIGENCE

41. Can your system work with Indian companies?

Answer: Yes. We propose extending the ingestion pipeline to **BSE/NSE filings and Indian company annual reports**. These documents can be processed, chunked, embedded and indexed using the same RAG architecture.

42. Why do you want Indian-market support?

Answer: The base-paper setup focuses on U.S. SEC filings. Indian-market support makes the system more relevant to Indian companies and users by incorporating BSE/NSE and Indian financial documents.

43. Why did you include Transaction Intelligence?

Answer: Financial RAG provides document-based intelligence, while Transaction Intelligence analyzes transaction behavior for anomalies and risk. Combining both provides a more complete financial investigation platform.

44. What is anomaly detection?

Answer: Anomaly detection identifies transactions or patterns that significantly differ from expected or normal behavior.

45. Will your system automatically block suspicious transactions?

Answer: No. Our system is designed as a **decision-support and investigation system**. It provides risk scores and alerts, while final actions remain subject to institutional policies and human review.

SECTION 10 — TECHNOLOGY, EVALUATION & SCOPE

46. What technologies are you planning to use?

Answer: The proposed stack includes **React + TypeScript + Tailwind CSS** for the frontend; **Python + FastAPI** for the backend; ML tools such as Pandas, NumPy, scikit-learn and XGBoost where appropriate; **LangChain** for RAG; **Docling** for document parsing; embeddings; **Qdrant** for vector retrieval; **BM25** for sparse retrieval; a reranker; an LLM; **Redis** for caching; Docker for deployment; and Git/GitHub for version control.

47. How will you evaluate your system?

Answer: We will evaluate **retrieval** using Recall@10, Recall@50, MRR and NDCG@10; **generation** using accuracy, faithfulness, completeness and hallucination rate; **system performance** using P95 latency, throughput and cache hit rate; and **user experience** using satisfaction and citation click-through rate.

48. What performance are you targeting?

Answer: Our project targets include **92%+ Recall@10, 95%+ Recall@50, 0.85+ MRR, 0.90+ NDCG@10, 85%+ answer accuracy, 95%+ faithfulness, less than 5% hallucination and less than 3 seconds P95 latency**. These are project targets, not experimentally achieved results yet.

49. What are the advantages and applications of your system?

Answer: Advantages include **evidence-grounded answers, traceable citations, hybrid retrieval, table-aware processing, multi-hop reasoning and unified document and transaction intelligence**. Applications include financial research, financial analysis, regulatory investigation, compliance, investor research and transaction-risk investigation.

50. What is the future scope of your project?

Answer: Future work includes **broader financial datasets, stronger multi-hop and agentic retrieval, improved real-time filing ingestion, wider BSE/NSE coverage, multilingual financial-document processing and scalable cloud deployment**.

EMERGENCY ANSWER

If the reviewer says: “Tell me about your project.”

“FinRAG AI is an integrated financial intelligence platform based on Retrieval-Augmented Generation. It retrieves relevant information from financial documents using metadata-aware, hybrid retrieval and reranking, and then uses an LLM to generate grounded answers with precise citations. We extend the base metadata-driven financial RAG approach with table-aware chunking, multi-hop reasoning, HyDE, real-time ingestion and Indian-market support through BSE/NSE sources. We also include a transaction-intelligence module for anomaly detection, risk scoring and alerts. The system is designed as a decision-support platform where final financial decisions remain with human experts.”

10 QUESTIONS TO MEMORIZE FIRST

1. What is your project?

FinRAG AI is an integrated financial intelligence platform using RAG for financial document question answering and transaction intelligence for risk analysis.

2. What is RAG?

Retrieval-Augmented Generation retrieves relevant external information and uses it as context for LLM generation.

3. What is your base paper?

Metadata-Driven Retrieval-Augmented Generation for Financial Question Answering.

4. Why did you select this base paper?

Because it directly addresses financial QA using metadata-driven RAG and provides a strong foundation for our proposed extensions.

5. What is the research gap?

Table handling, multi-hop reasoning, real-time ingestion, precise citations, Indian-market support and latency optimization.

6. What is your novelty?

Table-aware chunking, query decomposition/multi-hop reasoning, Indian sources, real-time ingestion, precise citations, HyDE and optimization.

7. How are you different from the base paper?

We extend it with table-aware processing, multi-hop reasoning, Indian-market support, real-time ingestion, precise citations and latency optimization.

8. Explain your methodology.

Collect → Parse → Chunk → Metadata → Embed → Index → Query → Hybrid Retrieve → Rerank → Generate → Cite → Evaluate.

9. Why table-aware and multi-hop retrieval?

Financial questions often depend on table structure and multiple pieces of evidence; preserving structure and decomposing complex queries improves grounded answering.

10. What accuracy have you achieved?

The base paper reports 88%. Our 90–92% figure is a project target, not an experimentally validated result yet.

CRITICAL REVIEW-1 REMINDER

Never present expected metrics as achieved results. If asked about your current accuracy, say: “Our target is 90–92%, but it has not yet been experimentally validated. We will report the actual measured result after implementation and evaluation.”